Exercise 7: Financial Forecasting

Scenario:

You are developing a financial forecasting tool that predicts future values based on past data.

Steps:

1. **Understand Recursive Algorithms:**
   - Explain the concept of recursion and how it can simplify certain problems.
2. **Setup:**
   - Create a method to calculate the future value using a recursive approach.
3. **Implementation:**
   - Implement a recursive algorithm to predict future values based on past growth rates.
4. **Analysis:**
   - Discuss the time complexity of your recursive algorithm.

Explain how to optimize the recursive solution to avoid excessive computation

<u>CODE:</u>

```
import java.util.Scanner;


public class FinancialForecast {


    public static double futureValue(double currentValue, double growthRate, int years) {


        if (years == 0) {
            return currentValue;
        }
```

```java
        return futureValue(currentValue * (1 + growthRate / 100), growthRate, years - 1);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter current value: ");
        double currentValue = sc.nextDouble();

        System.out.print("Enter annual growth rate (%): ");
        double growthRate = sc.nextDouble();

        System.out.print("Enter number of years: ");
        int years = sc.nextInt();

        double result = futureValue(currentValue, growthRate, years);

        System.out.printf("Future Value after %d years = %.2f", years, result);

        sc.close();
    }
}
```

<u>OUTPUT:</u>

```
Enter current value: 100000
Enter annual growth rate (%): 8
Enter number of years: 5
Future Value after 5 years = 146932.81
```